

## RESEARCH ARTICLE OPEN ACCESS

# Improving Implied Volatility Forecasts for American Options Using Neural Networks

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## ABSTRACT

This paper explores the application of neural networks to improve pricing of American options. Focusing on both American and European options on the S&P 100 index from January 2016 to August 2023, we integrate neural networks to model the difference between market-implied and model-implied volatilities derived from the Black-Scholes and Heston models. We also employ a pure neural network which does not use pricing models. Our study compares two neural network training architectures: independent training, where the network is trained anew daily, and sequential training, which uses the previous day's network setup as starting point to retrain the network. The findings reveal that a pure neural network with sequential training significantly reduces the root mean square error (RMSE) of implied volatility predictions compared to traditional pricing models. The inclusion of European option data in the training can enhance the models' forecasting performance, though its effectiveness varies depending on market conditions.

**JEL Classification:** G13, C45, C58

## 1 | Introduction

American options can be exercised at any time before expiration, unlike European options which can only be exercised at expiration. Accurate pricing of American options is crucial for determining their fair market value and for managing risk exposure. American options serve as an effective risk management tool, hedging against potential losses from other investments. Precise pricing methods can aid in making informed investment decisions and can significantly affect the profitability of trading strategies involving American options. Furthermore, investors may use American options to speculate on the future price movements of underlying assets, including stocks, commodities, and currencies.

However, due to market imperfections and theoretical model misspecification, mispricing of options and arbitrage opportunities emerge (Figlewski 1989; Constantinides et al. 2009). Figlewski (1989) underscores the mathematical complexities involved in integrating realistic market imperfections into

theoretical models. Additionally, Lindström et al. (2008) highlight that the mispricing of options is frequently attributed to the limited liquidity (high bid-ask spreads and low trading volumes) on organized exchanges. Li et al. (2024) find that illiquid options are significantly overpriced in the S&P 100 index options market. The scarcity of data contributes to inaccuracies in model calibration and mispricing that can result in financial losses. Therefore, it is imperative to employ appropriate models and efficiently use the available data to ensure the accurate pricing of American options.

Empirically, the traditional approach to pricing American and European options involves a two-step process: selecting a suitable mathematical model and then calibrating that model to fit real market data as accurately as possible (see, e.g., Fatone et al. 2024). It is common practice to invert option prices into implied volatilities that offer deep insights into market expectations of future price movements, making it an indispensable tool for formulating trading strategies and achieving a

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forward-looking view of the options market (Jeon et al. 2020). Since the seminal work of Black and Scholes (1973), a wide range of theoretical option pricing models have been developed (see Heston 1993; Cox et al. 1979). Recently, Lian et al. (2022) introduce a robust numerical method for pricing American options that incorporates a solvable algebraic equation for the early exercise boundary to ensure accurate and efficient option pricing. However, pricing accuracy depends not only on the choice of model but also on the quality of the calibration process: some information on the volatility surface might be lost because this process may not capture all of the nuances and complexities of the financial instruments.

Machine learning, particularly neural networks (NN), which are developed as universal approximators, offers promising avenues for advancing option pricing models. Hutchinson et al. (1994) were the first to propose the use of nonparametric methods and neural networks to estimate option prices. Since then, the application of neural networks to option pricing has undergone continuous development. Owing to their ability to model complex non-linear relationships and learn directly from vast datasets, neural networks can help to identify intricate patterns in market data and adapt to dynamic market conditions.

Neural networks have been widely applied in option pricing and hedging. Numerous studies focus on European options, showing that machine learning can significantly reduce computational demands (see Liu et al. 2019b; Ivaşcu 2021; De Spiegeleer et al. 2018; Ruf and Wang 2022). Horvath et al. (2021) and Liu et al. (2019a) demonstrate that neural networks effectively calibrate high-dimensional stochastic volatility models. Vuletić and Cont (2023) use generative adversarial networks to generate arbitrage-free implied volatility surfaces, while Ackerer et al. (2020) apply neural networks to enhance surface smoothness. Ruf and Wang (2020) provide a comprehensive review of the advancement of these techniques. Almeida et al. (2023) and Medvedev and Wang (2022) further improve the modeling of the S&P 500 implied volatility surface using feed-forward and recurrent neural networks, respectively.

In the American option pricing literature, Ait-Sahalia et al. (2010) show that the Heston model (Heston 1993) outperforms the Black-Scholes model (BS) (Black and Scholes 1973) for S&P 100, particularly for OTM contracts, due to its ability to capture early exercise features. Stentoft (2011) compares discrete and continuous-time models using a GARCH framework, while Gaspar et al. (2020) apply neural networks and benchmark them against the Least-Squares Monte Carlo method. Furthermore, daily forecasting of implied volatility is standard in practice (Almeida et al. 2023; Goncalves and Guidolin 2006; Bakshi et al. 1997). Real-time price adjustments help ensure accurate and timely option pricing in fast-moving markets.

Option prices on any given day are typically transformed into the implied volatility surface. This one-to-one mapping of prices to volatilities is valuable because implied volatilities provide a more consistent basis for comparison across different options than raw prices, both cross-sectionally and over time. Moreover, the shape of the volatility surface reflects the risk-neutral distribution of the underlying asset's returns, offering a clearer view of market expectations regarding future price movements. Institutional investors, in fact, routinely manage their option

portfolios using implied volatility (Carr and Wu 2016), which also serves as a cornerstone in the calibration of structural option pricing models (see Andersen et al. 2015; Ait-Sahalia et al. 2021). Consequently, precise modeling and forecasting of the implied volatility surface are essential for both academic research and trading performed in the industry and is in line with the literature.

In this paper, we propose a neural network model to capture the discrepancies between market-implied and model-implied volatilities for American options. We then use the estimated discrepancies to predict the implied volatility surface. Our novel methodology requires several steps, which are described below.

The first step is to convert American options into equivalent European options via a de-Americanization algorithm (see Burkovska et al. 2018; Carr and Wu 2010). This procedure enables us to use the equivalent European options for more straightforward model calibrations (see Merton 1973; Klemkosky and Resnick 1979). The second step is to fit parametric mathematical models, incorporating the Black-Scholes and Heston models into our analysis. In the third step, we employ a neural network to learn from the differences between model- and market-implied volatilities. Hence, we boost the Black-Scholes and Heston models with neural network. We also use a direct (pure) neural network to fit implied volatilities without the constraints of a parametric model. This helps to assess the neural network's ability to independently capture implied volatility dynamics, free from traditional model dependencies. We provide a comparative analysis of two neural network training architectures: one employing independent training, and the other a sequential training approach. The former independently trains a neural network each day to learn the features of the implied volatility, while the latter employs a sequential training architecture, learning from the model's weights and parameter values from the previous day. This comparison aims to ascertain the effectiveness of each training strategy in capturing the characteristics of the data.

Finally, we examine the predictive accuracy of our neural networks over different time horizons: same day, 1-day-ahead, and 5-days-ahead. For same day predictions, the data are split between training and testing sets either based on the options' strike price (K-splitting) or through random splitting (R-splitting), and the effects of these methods on performance are meticulously examined. Moreover, as our prediction setup includes forecasts extending to 1-day- and 5-days-ahead, we assess the robustness and reliability of our model for different forecast horizons.

In addition, we implement four prediction setups to evaluate whether incorporating S&P100 European option data improves the neural network's ability to predict S&P100 American option implied volatilities. These setups vary by training on European option data, American option data, or both, allowing for a systematic comparison of their predictive performance.

By investigating how these different models, neural network approaches, and configurations affect forecast accuracy, we make five key empirical contributions to the literature on American option pricing. First, we show that neural networks can enhance the implementation of traditional option pricing models such as Black-Scholes and Heston, by learning and correcting the discrepancies between model-implied and

market-implied volatilities. Second, we highlight the advantages of NN sequential training, showing that it consistently outperforms independent training by reducing the root mean square error (RMSE) of the forecasts and by adapting more effectively to evolving market conditions. Third, we find that the pure neural network (Pure NN) method, if sequentially trained, outperforms all other models. Fourth, we show that adding European options to American options to train the models and predict American options does not always help. Fifth, we suggest that all available options (as opposed to only options with volume larger than zero) should be included in the training set because they lead to a better fit. Together, these contributions demonstrate the potential of neural networks to improve the accuracy and robustness of option pricing frameworks; additionally, we provide a useful set of recommendations for the use of NN to predict American option implied volatilities.

The structure of the paper is as follows. Section 2 provides a description of the data. Section 3 outlines the model calibration process and the preparation of inputs for the neural network. Section 4 details the methodology, including the architecture of the feed-forward neural network and the independent and sequential training approaches. Section 5 presents the empirical results of predicting the implied volatility of American options. Finally, Section 6 concludes the paper.

## 2 | Data Description

Our data consist of S&P 100 options due to their status as the most liquid American index options available on OptionMetrics. Daily data on European and American S&P 100 index options are collected from OptionMetrics for the period spanning January 4, 2016, to August 31, 2023. OptionMetrics provides the highest closing bid and lowest closing ask quotes (across all US exchanges where the option trades), market-implied volatilities, strike prices, and times to maturity for each option contract. We use the midpoint of the bid and ask quotes as the option price. The market-implied volatilities are calculated by OptionMetrics using the midpoint of the bid-ask prices as the option market price, applying the inverse Black-Scholes model for European options and the binomial tree model for American options. Given that the S&P 100 index constituents regularly pay dividends, data on daily dividend yields are also sourced from OptionMetrics.

As proxy for risk-free rates, we use the zero-coupon yield curve obtained from OptionMetrics, matched to each option's time to maturity (interpolated when necessary). Dividend yields are also sourced from OptionMetrics and matched to each option's time to maturity. We source the daily closing prices of the S&P 100 index from Bloomberg, representing the underlying asset closing prices, on the same date as the option closing mid quote. The S&P 100 index price, the risk-free rate, and the dividend yield are all observed at a daily frequency. For a given day  $t$ , there are  $i = 1, \dots, n$  option observations, each associated with its corresponding option price, implied volatility, strike price, and time to maturity. All option observations share the same underlying index price  $S_t$ , while the risk-free rate  $r_{i,t}$  and the dividend yield  $q_{i,t}$  are matched to each option's time to maturity.

Following a common filtering choice adopted by the option pricing literature (see, for instance, Aït-Sahalia and Lo 1998;

Fan and Mancini 2009), we consider options with time to expiration ranging from 20 calendar days up to the maximum available maturity in the dataset (1094 days). Our sample includes options with moneyness between 0.8 and 1.6, covering in-the-money (ITM), at-the-money (ATM), and out-of-the-money (OTM) contracts, following the selection criteria in Almeida et al. (2023). Options are classified as ATM when moneyness lies between 0.97 and 1.03 for both calls and puts. For call options, contracts with moneyness between 1.03 and 1.6 are classified as in-the-money, while those with moneyness between 0.8 and 0.97 are classified as OTM. Conversely, for put options, contracts with moneyness between 1.03 and 1.6 are classified as OTM, whereas those with moneyness between 0.8 and 0.97 are classified as in-the-money. As for maturities, we classify options as short-term ([20, 240] days to expiration) and long-term ((240, 1094] days to expiration).

In summary, the original variables obtained from the OptionMetrics database include options' bid and ask quotes, market-implied volatilities, times to maturity, strike prices, underlying index prices, dividend yields, and risk-free rates. Our dataset comprises of 2,321,169 observations of European options and 2,572,366 observations of American options throughout the entire study period. On average, this equates to 1,205 European options and 1,338 American options per day. The dataset comprises of 1,160,021 call and 1,161,148 put records for European options, and 1,287,974 call and 1,284,392 put records for American options. When considering only actively traded contracts (volume > 0), the sample consists of 7,374 European options and 34,206 American options, yielding an average of 6 European options and 18 American options per day.

Since the OptionMetrics database provides implied volatilities derived from option quotes rather than transaction prices, our dataset includes both actively traded options (volume > 0) and non-traded options (volume = 0) on any given day. This approach is widely adopted in the literature on American options, where quote-based data are commonly used for pricing and implied volatility estimation (Ulrich et al. 2023; Cosma et al. 2020; Farias and Santa-Clara 2017; Carr and Wu 2010). Regardless of trading activity, the database consistently supplies all the necessary data for our analysis, such as strike prices, implied volatilities, time to maturity, and other relevant metrics. Notably, even on days when not many options are traded (particularly those with longer maturities or far OTM strikes), there can still be many valid quotes recorded reliably on OptionMetrics. Including zero-volume observations in our analysis not only increases the size of our dataset, which is advantageous for training our neural network model, but also helps us avoid bias toward more liquid options. This is appropriate since our focus is on implied volatility estimation rather than on liquidity effects. Furthermore, incorporating these additional observations allows us to achieve a fuller cross-section of strike prices and maturities, ultimately enhancing the robustness of our analysis.

Tables 1 and 2 provide a summary of the S&P 100 index implied volatility data for European and American options, respectively. The tables present the number of options, the mean implied volatility, and the standard deviation of the implied volatilities for each category of moneyness and time to maturity. Implied volatilities exhibit a clear skew with respect to moneyness, with higher values for OTM puts and ITM calls relative to ATM

**TABLE 1** | Summary statistics of S&P 100 European option market-implied volatilities.

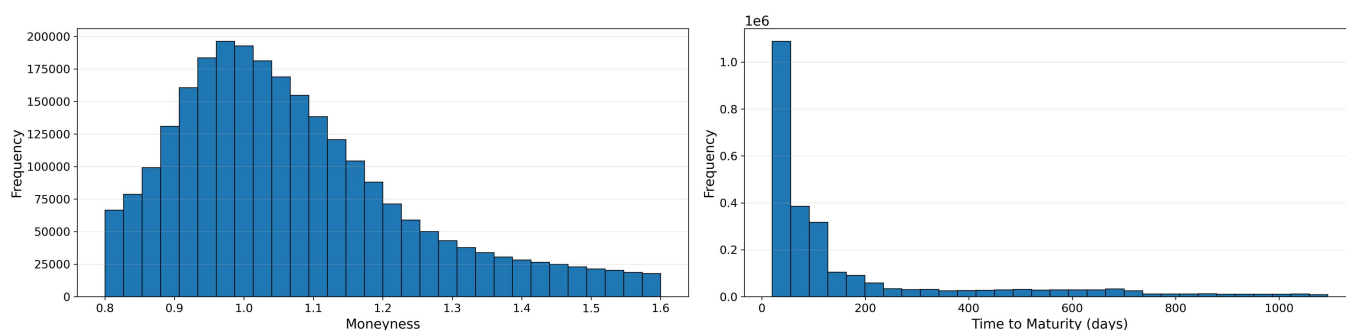
| Time to maturity |     | Observations |            | Mean     |            | Std. dev. |            |
|------------------|-----|--------------|------------|----------|------------|-----------|------------|
|                  |     | [20,240]     | [240,1094] | [20,240] | [240,1094] | [20,240]  | [240,1094] |
| Moneyness        |     |              |            |          |            |           |            |
| Call             | ITM | 501,815      | 165,633    | 0.2734   | 0.2457     | 0.0938    | 0.0499     |
|                  | ATM | 172,041      | 28,785     | 0.1658   | 0.1905     | 0.0636    | 0.0388     |
|                  | OTM | 211,417      | 80,330     | 0.1593   | 0.1585     | 0.0635    | 0.0414     |
| Put              | ITM | 280,386      | 89,066     | 0.1611   | 0.1568     | 0.0599    | 0.0379     |
|                  | ATM | 173,321      | 27,946     | 0.1655   | 0.1908     | 0.0629    | 0.0392     |
|                  | OTM | 443,626      | 146,803    | 0.2737   | 0.2458     | 0.0897    | 0.0494     |

Note: This table provides a summary of the market-implied volatilities for the S&P 100 European options, covering the time period from January 2016 to August 2023. The table reports the number of observations, the mean market-implied volatilities, and the standard deviation of the market-implied volatilities for varying levels of moneyness and time to maturity.

**TABLE 2** | Summary statistics of S&P 100 American option market-implied volatilities.

| Time to maturity |     | Observations |            | Mean     |            | Std. dev. |            |
|------------------|-----|--------------|------------|----------|------------|-----------|------------|
|                  |     | [20,240]     | [240,1094] | [20,240] | [240,1094] | [20,240]  | [240,1094] |
| Moneyness        |     |              |            |          |            |           |            |
| Call             | ITM | 557,144      | 167,171    | 0.3161   | 0.2513     | 0.1224    | 0.0532     |
|                  | ATM | 185,517      | 29,879     | 0.1720   | 0.1952     | 0.0637    | 0.0415     |
|                  | OTM | 272,904      | 75,359     | 0.1655   | 0.1616     | 0.0620    | 0.0478     |
| Put              | ITM | 353,598      | 91,066     | 0.2039   | 0.1715     | 0.0722    | 0.0365     |
|                  | ATM | 186,397      | 29,142     | 0.1729   | 0.2000     | 0.0621    | 0.0415     |
|                  | OTM | 495,107      | 129,082    | 0.2781   | 0.2452     | 0.0919    | 0.0490     |

Note: This table provides a summary of the market-implied volatilities for the S&P 100 American options, covering the time period from January 2016 to August 2023. The table reports the number of observations, the mean market-implied volatilities, and the standard deviation of the market-implied volatilities for varying levels of moneyness and time to maturity.

**FIGURE 1** | Histogram by time to maturity and moneyness For American options. The left figure depicts the histogram of moneyness, while the right figure illustrates the histogram of time to maturity of our sample, spanning from January 2016 to August 2023.

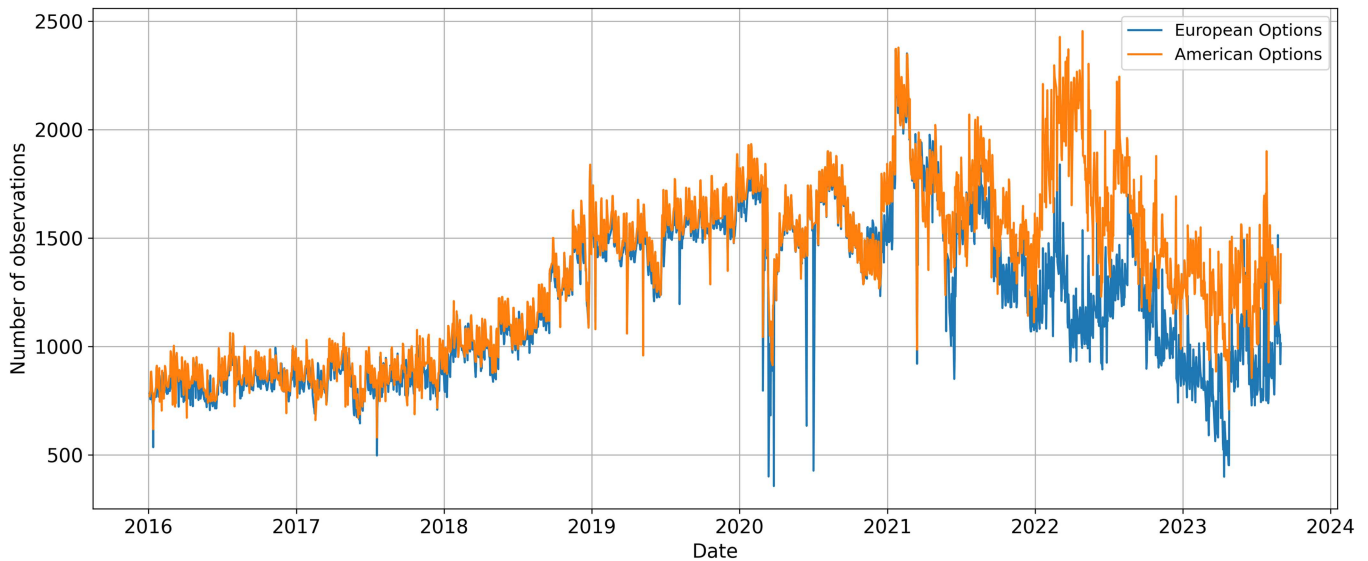
options for both option types. American options display systematically higher implied volatilities than European options, particularly for shorter maturity options, reflecting the value of early-exercise flexibility and dividend effects. In addition, the dispersion of implied volatility decreases with maturity, as indicated by lower standard deviations for longer-dated options. Overall, the observed patterns are consistent with established empirical properties of equity index option markets.

Figure 1 illustrates the distributions of moneyness and time to maturity (in days) for American options in the sample. The maturity distribution is highly right-skewed, with a large concentration of shorter-dated options, particularly those with

maturities below 100 days, while the number of observations declines steadily as time to maturity increases. The moneyness distribution is centered around one, indicating that ATM options constitute the largest share of observations. ITM and OTM options become increasingly sparse toward the tails. Overall, the distributions reflect typical trading patterns in equity index option markets, where liquidity is concentrated in shorter-term and near-the-money contracts.

Figure 2 illustrates the number of daily observations of European and American options. It is evident that American options have consistently higher daily observation counts than European options throughout the study period. From January





**FIGURE 2** | The frequency of observations for American and European options across time. This figure illustrates the number of observations for both European and American options over a period extending from 2016 to 2023. European options are depicted in blue, while American options are represented in orange.

2016 to December 2019, the daily observations for both European and American options predominantly range between 500 and 2000. Subsequently, observation counts for both European and American options exhibit pronounced drops around early 2020, corresponding to the onset of the COVID-19 market disruption, followed by a sharp rebound and heightened volatility during the rest of 2020 and 2021. After peaking in 2021 and 2022, the number of observations for both option types declines toward the end of the sample period, although American options remain more frequently quoted than European options.

### 3 | Model-Implied Volatility Estimation

In this section, we explain the calibration of both the Black-Scholes and Heston models for European and American options. We also describe the estimation of model-implied volatilities for both types of options under these models.

#### 3.1 | European Options

The estimation of European option prices is considered standard in the literature. In the following, we review the two most popular models.

##### 3.1.1 | Black-Scholes Model for European Options

The seminal Black-Scholes model (Black and Scholes 1973) is based on the assumption that the underlying asset price follows a Geometric Brownian Motion process:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where  $S_t$  is the underlying asset price at time  $t$ ,  $\mu$  is expected asset return,  $\sigma$  is the constant volatility, and  $W_t$  is a Wiener process. The price of a European call option  $C_{BS}(S_t, K, \tau, r, q, \sigma_E)$  is:

$$C_{BS}(S_t, K, \tau, r, q, \sigma) = S_t e^{-q\tau} \Phi(d_1) - K e^{-r\tau} \Phi(d_2).$$

Here  $K$  is the strike price of the option,  $r$  is the risk-free interest rate,  $q$  is the dividend yield,  $\tau = T - t$  is the time to expiration,  $\Phi(\cdot)$  is the cumulative distribution function of the standard normal distribution and  $d_1$  and  $d_2$  are given by:

$$d_1 = \frac{\log\left(\frac{S}{K}\right) + \left(r - q + \frac{\sigma^2}{2}\right)\tau}{\sigma\sqrt{\tau}}, d_2 = d_1 - \sigma\sqrt{\tau}.$$

Similarly, the price of a European put option is given by:

$$P_{BS}(S_t, K, \tau, r, q, \sigma) = K e^{-r\tau} \Phi(-d_2) - S_t e^{-q\tau} \Phi(-d_1).$$

For each trading day  $t$ , the Black-Scholes model is calibrated to observed European option prices by estimating a single constant volatility parameter, denoted by  $\sigma_{BS, Eu}$ . Let  $O_{i,t}$  denote the observed market price of the  $i^{\text{th}}$  European option (call or put) with strike price  $K_i$  and time to maturity  $\tau_i$ , observed on day  $t$ . The calibration is performed by minimizing the mean squared pricing error between market prices and Black-Scholes model prices across all  $n$  available options on that day.

Formally, the calibration problem is defined as:

$$\min_{\sigma_{BS, Eu}} \frac{1}{n} \sum_{i=1}^n [O_{i,t} - O_{BS}(S_t, K_i, \tau_i, r_{i,t}, q_{i,t}, \sigma_{BS, Eu})]^2, \quad (1)$$

where  $O_{BS}(\cdot)$  denotes the Black-Scholes price of a European call or put option, depending on the contract type, evaluated using the constant volatility parameter  $\sigma_{BS, Eu}$ . The inputs include the underlying index price  $S_t$ , strike price  $K_i$ , option maturity  $\tau_i$ , maturity-matched risk-free rate  $r_{i,t}$ , and dividend yield  $q_{i,t}$ .

The Black-Scholes model assumes a constant implied volatility  $\sigma_{BS, Eu}$  across all strike prices and maturities. However, empirical studies on the implied volatility surface consistently reveal a pronounced 'skew' or 'smile' pattern across different strike prices and maturities as also detailed in Bates (1991), Pan (2002) and Doran et al. (2007). This leads to systematic mispricings by the Black-Scholes model, which fails to capture the complexities of market behavior observed in real-world data.

### 3.1.2 | Heston Model for European Options

This section provides a summary description of the calibration of the Heston model and the estimation of its model-implied volatility for European options. The Heston model is an extension of the Black-Scholes model allowing for stochastic volatility with mean reversion, which is a more realistic assumption in financial markets (Heston 1993).

The dynamics for the underlying asset is given by the process below:

$$dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^1,$$

$$dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t}dW_t^2,$$

Here  $S_t$  is the stock price at time  $t$ ,  $v_t$  is the instantaneous variance of the stock price,  $W_t^1$  and  $W_t^2$  are two correlated standard Brownian motions,  $\kappa$  is the mean reversion rate of the variance,  $\theta$  is the long-term variance level,  $\xi$  is the volatility of the variance and the correlation between  $dW_t^1$  and  $dW_t^2$  is  $\rho$ .

In the calibration of the Heston model for European options, we aim to estimate the model parameters by minimizing the sum of squared differences between the observed market prices of options and the prices obtained using the Heston model. This process is conducted separately for call and put options as detailed below.

For each trading day  $t$ , the minimization we perform for options is given by:

$$\min_{\Pi} \frac{1}{n} \sum_{i=1}^n [O_i - O_{\text{Heston, Eu}}(S_t, K_i, r, q, \tau_i, \Pi)]^2, \quad (2)$$

where  $O_i$  represents the observed market price of the  $i$ -th option (either call or put),  $O_{\text{Heston, Eu}}$  denotes the model price computed using the Heston model for European options (calls and puts), and  $\Pi = (\theta, \kappa, \xi, \rho)$  is the set of Heston model parameters to be calibrated.

The optimization is conducted separately for calls and puts. For call options, the minimization leads to a calibrated parameter set  $\hat{\Pi}_c$ . Similarly, for put options, we obtain a separate parameter set  $\hat{\Pi}_p$ .

We use the calibrated parameter sets  $\hat{\Pi}_c$  and  $\hat{\Pi}_p$  for call and put options on each trading day  $t$  to compute the corresponding Heston model option prices. Specifically, we apply a quasi-closed-form solution using the fast Fourier transform (FFT) method, as described by Carr and Madan (1999). This approach relies on the characteristic function of the log returns of the underlying asset and involves transforming the option payoff

function into the frequency domain via the Fourier transform. The option price is then recovered by applying the inverse Fourier transform.

Using this method, we obtain the Heston model prices for calls and puts, denoted by  $C_{\text{Heston, Eu}}$  and  $P_{\text{Heston, Eu}}$ , respectively, for each option on day  $t$ . To derive the model-implied volatilities from these prices, we apply the inverse Black-Scholes pricing formula, defined separately for call and put options as follows:

$$\sigma_{\text{Heston, Eu}, i} = \begin{cases} C_{BS}^{-1}(C_{\text{Heston, Eu}, i}, S_t, K_i, r, q, \tau_i, \hat{\Pi}_c) & \text{for a call option} \\ P_{BS}^{-1}(P_{\text{Heston, Eu}, i}, S_t, K_i, r, q, \tau_i, \hat{\Pi}_p) & \text{for a put option} \end{cases}.$$

Here,  $C_{BS}^{-1}$  and  $P_{BS}^{-1}$  represent the inverse Black-Scholes formulas used to back out implied volatility from the given option prices. The inputs include the underlying price  $S_t$ , strike price  $K_i$ , time to maturity  $\tau_i$ , risk-free rate  $r$ , and dividend yield  $q$ . To solve for the implied volatility, we employ the Brent method (Brent 1971), a robust and efficient numerical root-finding algorithm that iteratively inverts the Black-Scholes formula.

## 3.2 | American Options

This section describes the procedure to obtain model-implied volatilities for American options. We start with describing the de-Americanization process for American options. This is followed by providing the details on obtaining model-implied volatilities using the Black-Scholes model and Heston model for American options.

### 3.2.1 | De-Americanization

De-Americanization is the process of transforming American option prices into equivalent European option prices. American option pricing takes into account the early exercise feature, which allows the option holder to exercise the option before expiration. In the following, we use the de-Americanization algorithm of Carr and Wu (2010) to convert American options into equivalent European options. Following Carr and Wu (2010), we use out-of-the-money options to minimize approximation errors. The de-Americanization method has been shown to be both efficient and easy to implement for calibrating American options (Burkovska et al. 2018), and has been applied in recent studies such as Ouazad (2022) to calibrate options on individual stocks.

We convert American option prices by first converting them into implied volatilities ( $\sigma_{Am}$ ) using the binomial tree model, following the approach of Cox et al. (1979). This methodology is consistent with the procedure adopted by the OptionMetrics database. Subsequently, following the de-Americanization method proposed by Carr and Wu (2010), these implied volatilities are used within the Black-Scholes framework to compute the equivalent European option prices:

$$C_{A\_equivalent} = S_0 e^{-q\tau} N(d_1) - Ke^{-r\tau} N(d_2), \quad (3)$$

$$P_{A\_equivalent} = Ke^{-r\tau} N(-d_2) - S_0 e^{-q\tau} N(-d_1), \quad (4)$$

$$d_1 = \frac{\log\left(\frac{S_0}{K}\right) + \left(r - q + \frac{\sigma_{Am}^2}{2}\right)\tau}{\sigma_{Am} \sqrt{\tau}}, \quad d_2 = d_1 - \sigma_{Am} \sqrt{\tau}.$$

This process effectively treats the market-implied volatilities of American options as if they are derived from European options, allowing us to obtain the corresponding European-equivalent option prices. In the following, we use these de-Americanized prices when using the Heston model with neural networks for American options.

### 3.2.2 | Black-Scholes Model for American Options

Using the de-Americanized European-equivalent option prices obtained from Equations (3) and (4), we calibrate the Black-Scholes model for American options by estimating a single constant volatility parameter, denoted by  $\sigma_{BS,Am}$ , on each trading day  $t$ .

Let  $O_{A\_equivalent,i}$  denote the de-Americanized market price of the  $i^{\text{th}}$  American option with strike price  $K_i$  and time to maturity  $\tau_i$ . The calibration is performed by minimizing the mean squared pricing error between the de-Americanized option prices and the corresponding Black-Scholes model prices across all  $n$  available options on day  $t$ :

$$\min_{\sigma_{BS,Am}} \frac{1}{n} \sum_{i=1}^n [O_{A\_equivalent,i} - O_{BS}(S_t, K_i, \tau_i, r_{i,t}, q_{i,t}, \sigma_{BS,Am})]^2, \quad (5)$$

where  $O_{BS}(\cdot)$  denotes the Black-Scholes price of a European call or put option, depending on the contract type, evaluated using the constant volatility parameter  $\sigma_{BS,Am}$ . The inputs include the underlying index price  $S_t$ , strike price  $K_i$ , maturity  $\tau_i$ , and maturity-matched risk-free rate  $r_{i,t}$  and dividend yield  $q_{i,t}$ .

After calibration, Black-Scholes model-implied volatilities for American options are obtained by inverting the Black-Scholes pricing formula using the calibrated volatility parameter  $\sigma_{BS,Am}$ .

### 3.2.3 | Heston Model for American Options

Using the prices  $C_{A\_equivalent}$  and  $P_{A\_equivalent}$  obtained in Equations (3) and (4) through the de-Americanization process, we calibrate the Heston model parameters separately for call and put options by minimizing the mean squared differences between the de-Americanized option prices and the model-generated Heston prices. The optimization problem is given by:

$$\min_{\Pi} \frac{1}{n} \sum_{i=1}^n [O_{A\_equivalent,i} - O_{Heston,A\_equivalent}(S_t, K_i, r, q, \tau_i, \Pi)]^2, \quad (6)$$

where  $O_{A\_equivalent,i}$  denotes the de-Americanized market price of the  $i^{\text{th}}$  option (either call or put), and  $O_{Heston,A\_equivalent}$  denotes the corresponding Heston model price. Here,  $\hat{\Pi} = (\theta, \kappa, \xi, \rho)$  denotes the set of Heston model parameters calibrated on day  $t$  for American options. The calibration is performed separately for calls and puts. For call options, the minimization leads to the parameter set  $\hat{\Pi}_c$ . For put options, we obtain a separate parameter set  $\hat{\Pi}_p$ . Once calibration is complete, we use the estimated parameter set to compute the Heston model-based prices for American options, denoted as  $C_{Heston,A\_equivalent}$  and  $P_{Heston,A\_equivalent}$ , respectively.

Finally, to obtain the corresponding Heston model-implied volatilities for American options, we invert the model-generated prices using the Black-Scholes formula:

$$\sigma_{Heston,Am,i} = \begin{cases} C_{BS}^{-1}(C_{Heston,A\_equivalent,i}, S_t, K_i, r, q, \tau_i, \hat{\Pi}_c) & \text{for a call option,} \\ P_{BS}^{-1}(P_{Heston,A\_equivalent,i}, S_t, K_i, r, q, \tau_i, \hat{\Pi}_p) & \text{for a put option.} \end{cases}$$

## 4 | American Option Implied Volatility Forecasting Using Neural Networks

In this section, we discuss the objective function that our neural network is designed to learn. We then explain the data splitting strategies employed for training and testing. This is followed by a detailed description of the forecasting setups and the structure of the neural networks. Finally, we present two types of training approach that we employ: independent training and sequential training, applied to both the Black-Scholes model and the Heston model.

### 4.1 | Implementation of Neural Network Models—Overview

The neural network frameworks we use to forecast American option implied volatilities follow the methodology proposed by Almeida et al. (2023) and related literature, adapting the procedure to the American option setting considered in this study.

We consider two different neural network frameworks. First, we implement models boosted by neural networks, where the neural networks are trained to learn the implied volatility error surfaces between market-implied volatilities and model-implied volatilities obtained from parametric pricing models. These models include the Black-Scholes and Heston models. Second, we implement a pure neural network approach, in which the network learns the implied volatility surface directly from the data without relying on any parametric pricing model.

#### 4.1.1 | Models Boosted With Neural Network

In the following, we present the neural network framework that boosts parametric option pricing models for implied volatility forecasting. Let  $p$  denote the underlying pricing model, where  $p \in \{BS, Heston\}$ . Each model  $p$  is first calibrated to option price data, yielding a model-implied volatility surface  $\sigma_p(m, \tau)$  with  $m$  and  $\tau$  denoting moneyness and maturity.

Let  $\sigma_{Market}(m, \tau)$  denote the market-implied volatility. We define the implied volatility error surface associated with model  $p$  as:

$$\epsilon_p(m, \tau) = \sigma_{Market}(m, \tau) - \sigma_p(m, \tau).$$

We assume that the error surface  $\epsilon_p(m, \tau)$  is a smooth function of moneyness  $m = S/K$  and time to maturity  $\tau$ . On each trading day  $t$ , a neural network is trained to approximate this error surface by solving the following minimization problem:

$$\min_{f_p} \frac{1}{n} \sum_{i=1}^n [\epsilon_p(m_{i,t}, \tau_{i,t}) - f_p(m_{i,t}, \tau_{i,t})]^2, \quad (7)$$

where  $m_{i,t}$  and  $\tau_{i,t}$  denote the moneyness and time to maturity of option  $i$  observed on day  $t$ , and  $f_p(\cdot)$  is the function approximated by the neural network.

Given the estimated correction function  $\hat{f}_p(\cdot)$ , the implied volatility forecast for an out-of-sample option price for day  $t + h$ , for option  $j$ , is constructed as:

$$\hat{\sigma}_{p,NN}(m_{j,t+h}, \tau_{j,t+h}) = \sigma_p(m_{j,t+h}, \tau_{j,t+h}) + \hat{f}_p(m_{j,t+h}, \tau_{j,t+h}), \quad (8)$$

where  $h \geq 0$  denotes the forecast horizon.

#### 4.1.2 | Pure Neural Network

Next, we illustrate the pure neural network framework that directly learns the implied volatility surface without relying on any parametric option pricing model.

Let  $\sigma_{\text{Market}}(m, \tau)$  denote the market-implied volatility. For each trading day  $t$ , we assume that the implied volatility surface can be represented by an unknown function  $g(m, \tau)$ . Given  $n$  in-sample option price observations on day  $t$ , the neural network is trained by minimizing the following objective function:

$$\min_g \frac{1}{n} \sum_{i=1}^n [\sigma_{\text{Market}}(m_{i,t}, \tau_{i,t}) - g(m_{i,t}, \tau_{i,t})]^2, \quad (9)$$

where  $m_{i,t}$  and  $\tau_{i,t}$  denote the moneyness and time to maturity of option  $i$  observed on day  $t$ .

The estimated function  $\hat{g}(\cdot)$  provides a flexible, nonlinear approximation of the implied volatility surface. For out-of-sample option prices on day  $t + h$ , the implied volatility forecast is obtained as:

$$\hat{\sigma}_{NN}(m_{j,t+h}, \tau_{j,t+h}) = \hat{g}(m_{j,t+h}, \tau_{j,t+h}), \quad (10)$$

where  $h \geq 0$  denotes the forecast horizon and  $j$  denotes the option for which we forecast the implied volatility.

Therefore, Pure NN does not have a model to rely on and uses only implied volatilities as input. The models boosted with neural network (Model-enhanced NN), on the other hand, use the models' information, and a well-established way to do this is to have the difference between the model-implied and market-implied volatilities as the input to the NN—see Almeida et al. (2023). This means that the inputs are somewhat different, but the input of the Model-enhanced NN (difference between model- and market-implied volatilities) is derived from the same input as the Pure NN (market implied volatilities). Hence, using a model, if well specified, could help the NN to “learn”; however, we do not know a priori whether the Model-enhanced NN will perform better than a Pure NN. We set a “race” between these different NN approaches which allows us to jointly test the model and the NN performance.

It is important to clarify that, given that the Black-Scholes model implied volatility is a constant, the minimization problems outlined in Equations (7) and (8) and (9) and (10) are mathematically equivalent, under an independent training approach (see Section 4.5). We note that this equivalence does not hold under sequential training, where the neural network

parameters are updated using the previous day's optimal values, and the Pure NN and Black-Scholes model boosted with neural network capture the time variation in different ways, as explained in Section 4.5. Due to the equivalence, in the empirical analysis we report the results for the Black-Scholes and Heston models boosted with NN both independently and sequentially trained, and Pure NN only sequentially trained.

## 4.2 | Data Splitting Into Training and Testing Subsets

We explore two distinct approaches to split data into training and testing subsets for same-day prediction tasks. The first method referred to as “K-splitting” in our paper, follows the methodology introduced by Almeida et al. (2023). For each day  $t$  within our sample, we divide the option data into training and testing sets based on the divisibility of the strike price. Specifically, the models are trained on options with strike prices divisible by 10, while testing is conducted on options with strikes that are not divisible by 10. This method assumes that multiple option contracts are available at each strike price, which allows for a systematic segmentation based on this numerical property. In practical trading scenarios, one often encounters situations where only a subset of the full cross-section of option data is available on any given trading day  $t$ , and the goal is to leverage these limited data to accurately predict the prices of other unobserved options within the same trading session. The “K-splitting” method effectively mimics the challenge of making predictions based on partial strike price data.

The second method, which we call “R-splitting,” employs random splitting which is a commonly used technique in machine learning. This approach uses a stochastic process to divide the data into training and testing subsets, adhering to a 70:30 division ratio. Consequently, approximately 70% of the data are used for training the model, with the remaining 30% reserved for testing purposes. Random splitting ensures a heterogeneous mix of option data across both sets, promoting a diverse representation that enhances the robustness and generalizability of the predictive models. “R-splitting” represents a scenario where the available data is selected randomly. This method reflects the unpredictable nature of real-world data, providing a robust test of the model's ability to generalize from a randomly selected subset to the broader market.

For both 1-day- and 5-days-ahead predictions, we use the complete dataset on day  $t$  to train the model. The test set comprises data on day  $t + h$ , where  $h$  equals 1 for 1-day-ahead predictions and 5 for 5-day-ahead predictions. This approach ensures that the model is trained on current information and then evaluated on its ability to forecast future market conditions based on data from subsequent days.

## 4.3 | Using European Option Data to Predict American Option Implied Volatility

Considering the distinct characteristics of European and American options, particularly their exercise traits, in the following we examine the extent to which data on European options, in addition to data on American option prices, can



inform and improve the predictive accuracy of pricing models for American option implied volatility.

To address this question, we undertake four distinct prediction setups that we consistently implement across each day  $t$  and for each data splitting methodology employed in our study. The prediction setups are designed to evaluate various combinations of training and testing datasets comprising European and American option data. The first prediction setup (EU to EU) uses European option data exclusively for both the training and testing phases to predict European option implied volatility. The second prediction setup (AM to AM) employs American option data to train a feed-forward neural network and subsequently predict American option implied volatility within the test dataset. The third prediction setup (EU to AM) leverages European option data as the training set with the objective of predicting American option implied volatility in the test set. Finally, the fourth prediction setup (EU + AM to AM) integrates both European and American option data for model training, followed by testing the model's performance to predict American option prices. This multifaceted approach allows for a comprehensive evaluation of the predictive performance across different option types and training scenarios.

For each of the last three prediction configurations, the testing dataset for American options remains constant, ensuring a consistent comparison. Similarly, the training datasets for both European and American options are aligned in terms of data splitting methodology, thereby facilitating a comprehensive analysis of the added value of combining European and American option data in training predictive models. By examining these distinct configurations, we aim to elucidate the potential synergies between European and American option data in enhancing the accuracy and robustness of option pricing models.

#### 4.4 | Structure of Feed-Forward Neural Networks

The neural network employed in this paper includes an input layer, multiple hidden layers, and an output layer. The model progresses through  $L$  hidden layers, where each layer  $l$  transforms its input through a combination of weighted sums and a nonlinear activation function. The architecture is defined as follows:

$$z_0 = x_{i,t}$$

$$z_l = w^o(A_{l-1}z_{l-1} + b_{l-1}), \text{ for } l = 1, 2, \dots, L$$

$$f(x_{i,t}) = A_L z_L + b_L.$$

The input layer receives variables  $x_{i,t} = (m_{i,t}, \tau_{i,t})'$  for each option  $i$  on day  $t$ , where the variables represent the moneyness and time to maturity.  $A_{l-1}$  denotes the transformation matrix for layer  $l - 1$ .  $b_{l-1}$  is the bias vector for layer  $l - 1$ .  $w^o$  is the activation function applied element-wise, typically employing functions such as the sigmoid or the ReLU (Rectified Linear Unit).

As detailed in Table 3, our neural network architecture is composed of three hidden layers with 32, 16, and 8 neurons,

respectively. The ReLU function is employed as the activation function across all hidden layers. The network is trained using the Adam optimizer for a total of 800 epochs, with a batch size of 32. This configuration is designed to exploit the depth and non-linearity of neural networks to effectively capture and model the intricate patterns inherent in option pricing data.

#### 4.5 | Independent Versus Sequential Training Approaches

In this section, we describe two training approaches that we implement and compare in this study. The independent training approach resets the neural network's weights and biases daily, meaning that each day's training starts from a randomly initialized model without any prior knowledge. In contrast, the sequential training approach retains and updates the weights and biases over time, allowing the model to continuously learn from past training data while incorporating new information.

##### 4.5.1 | Independent Training Approach

In the independent training approach, for each day  $t$ , we set the matrix of initial weights  $W_{l,t}$  and biases vector  $b_{l,t}$  for each layer of the feed-forward neural network to new, random values:

$$W_{l,t}^{(new)}, b_{l,t}^{(new)} \quad \text{for } l = 0, 1, \dots, L.$$

Here,  $L$  is the total number of layers in the feed-forward neural network,  $l$  is the index for the layer, and  $D_t$  is the training data available on day  $t$ . Next, we train the model. Specifically, we use training data on day  $t$  to update the weights and biases:

$$(W_{l,t}, b_{l,t}) = \text{train}\left((W_{l,t}^{(new)}, b_{l,t}^{(new)}), D_t\right).$$

##### 4.5.2 | Sequential Training Approach

In the sequential training approach, the weights of the neural network are initialized once and then updated each day based on new data. This can be mathematically represented as follows:

At the beginning of the training process, the weights  $W$  and biases  $b$  for each layer are initialized:

**TABLE 3** | Summary of the feed-forward neural network.

| Layer (Type)          | Size     | Parameters |
|-----------------------|----------|------------|
| Hidden-Layer1 (Dense) | (32, 32) | 96         |
| Hidden-Layer2 (Dense) | (32, 16) | 528        |
| Hidden-Layer3 (Dense) | (32, 8)  | 136        |
| Output-Layer (Dense)  | (32, 1)  | 9          |

*Note:* This table summarizes the architecture of the feed-forward neural network used in the study. It details the types and output shapes of each layer, along with the number of parameters for each layer. This includes multiple dense layers leading up to the output layer, which together facilitate the learning and prediction tasks of the network. The term 'Parameters' denotes the total number of trainable parameters within each layer, encompassing both weights and biases. The output size  $(m, n)$  indicates  $m$  batches and  $n$  neurons, where  $m$  represents the batch size and  $n$  reflects the number of output features from each layer.

$$W_l^{(\text{init})}, b_l^{(\text{init})} \quad \text{for } l = 0, 1, \dots, L.$$

Here  $W_l, b_l$  represent the matrix of weights and biases vector for layer  $l$ .  $D_t$  represents the training data available on day  $t$ , and the function `train()` adjusts these parameters to minimize the loss on data on that day, improving the model based on both the new data and what has been learned in the previous days.

Next, for each day  $t$ , we sequentially train the model:

$$(W_{l,t}, b_{l,t}) = \text{train}((W_{l,t-1}, b_{l,t-1}), D_t).$$

Unlike in the independent approach, here  $W_{l,t}$  and  $b_{l,t}$  are updated daily without re-initialization, allowing the model to learn incrementally and potentially recognize more complex patterns over time.

It is important to clarify that under the independent training approach, the Black-Scholes model boosted with neural network is equivalent to the pure neural network, as also noted in Almeida et al. (2023). The minimization problems outlined in Equations (7) and (8) and (9) and (10) are mathematically equivalent given that the Black-Scholes model implied volatility  $\sigma_{BS}$  is a constant. However, this equivalence does not hold under sequential training, where the neural network parameters are updated over time and the pure neural network and Black-Scholes model boosted with neural network capture the time variation in different ways. Accordingly, in the empirical analysis we report the results for the Black-Scholes and Heston models boosted with neural network both independently and sequentially trained, and the pure neural network only sequentially trained.

As explained before in Section 4.1.2, the Pure NN uses as input the implied volatility surface, whereas the BS + NN approach uses as input the deviation from a flat Black-Scholes surface. Also, the Pure NN and the Black-Scholes model enhanced with neural network formulations are mathematically equivalent when trained independently on a given day. However, this equivalence does not hold under sequential training. In the sequential training setup, the parameters of NN are initialized with the optimal values of the previous day. So, in this setting we use additional information (parameter values from the previous day) which can boost the performance of the NN. Given that the setups of the Pure NN and BS + NN are different (one uses volatilities directly as input and the other used volatility deviations as input), the structure of the hyperparameters is also different. This makes the variation in time in the parameters be propagated differently in these two setups and causes a difference in their performance. Consequently, due to the complex nature of neural networks, equivalence between Pure NN and BS + NN model holds under independent training but does not hold under sequential learning.

## 5 | Empirical Results

In the following, we present our results. We begin with in-sample results for the same day, 1-day-ahead, and 5-days-ahead predictions, followed with out-of-sample results.<sup>1</sup> Finally, we conduct a robustness analysis where we use only actively traded options to train the neural network. For each forecasting

horizon, we evaluate the performance of different prediction setups and data splitting methods. The analysis focuses on the ability of a model to fit (in-sample) and forecast (out-of-sample) the implied volatility surface, using RMSE (based on the difference between market-implied volatilities and model-implied volatility forecasts/fit) to assess both in-sample and out-of-sample accuracy. In particular, the RMSE of the models is computed as follows, with  $\sigma_{\text{Market}}$  the market-implied volatility and  $\hat{\sigma}$  the out-of-sample model-implied volatility forecast for day  $t$  (or, for in-sample RMSE, the fitted model-implied volatility), computed using various models and setups:

$$\text{RMSE} = \sqrt{\frac{1}{m} \sum_{i=1}^m [\sigma_{\text{Market}}(m_{i,t}, \tau_{i,t}) - \hat{\sigma}(m_{i,t}, \tau_{i,t})]^2}.$$

### 5.1 | In-Sample Results

For same day predictions, we implement the R-splitting and K-splitting methods. For 1-day- and 5-days-ahead forecasts, the models are trained on the same in-sample dataset. Table 4 reports in-sample RMSE for same day (Panels A and B), 1-day-ahead, and 5-days-ahead (Panel C) implied volatility fits under different model specifications, data-splitting methods, and training approaches.<sup>2</sup> Several clear patterns emerge. First, the parametric Black-Scholes and Heston models produce substantially larger in-sample errors than all neural network-based specifications, reflecting their limited ability to capture the shape of the implied volatility surface. Second, boosting either pricing model with neural networks substantially reduces RMSE, with sequential training generally improving fit relative to independent training. Third, the sequentially trained Pure NN delivers the lowest in-sample RMSE across all three panels and data-splitting methods in general, particularly for American option predictions. Finally, the ranking of model performance is broadly consistent across K-splitting and R-splitting and across same-day and multi-days-ahead in-sample fits, suggesting that these improvements are not driven by a specific data-splitting method.

### 5.2 | Out-of-Sample Results

Table 5 reports the out-of-sample RMSE for same day, 1-day- and 5-days-ahead predictions. Figures 3 and 4 display the 60-day moving average of the out-of-sample RMSE for same day predictions using K-splitting and R-splitting, respectively. For comparison, these figures present the RMSE for the BS and Heston models without neural networks, the models boosted with neural networks (both independently and sequentially trained), and for the sequentially trained Pure neural network. Figures 5 and 6 present the 60-day moving average of the out-of-sample RMSE for same day predictions, separately for the “AM to AM” and “(EU + AM) to AM” prediction setups, using the BS + NN, Heston+NN, and Pure NN models with sequential training. Figure 5 presents the results with K-splitting, while Figure 6 those with R-splitting. In the following, we discuss and interpret the results from different perspectives.

As can be seen from Table 5, the models enhanced with neural networks (BS + NN and Heston + NN specifications) consistently

**TABLE 4** | In-sample RMSE.

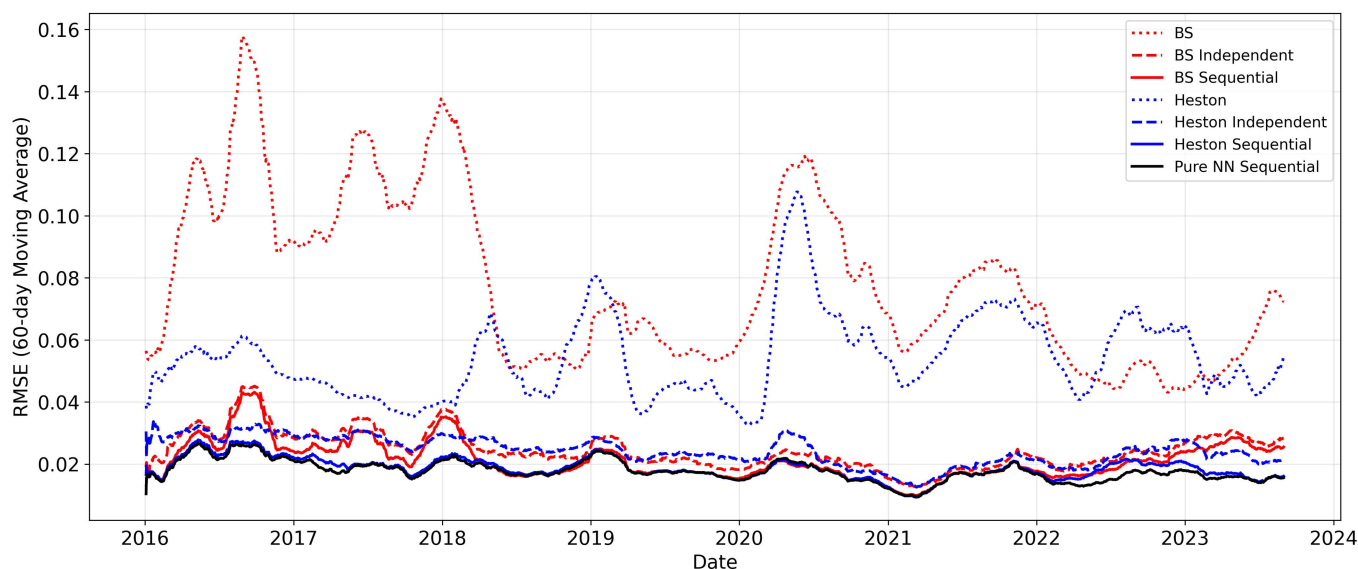
|                                  | BS     | Heston | BS+NN       |            | Heston+NN   |            | Pure NN<br>Sequential |
|----------------------------------|--------|--------|-------------|------------|-------------|------------|-----------------------|
|                                  |        |        | Independent | Sequential | Independent | Sequential |                       |
| Panel A: Same day (K-splitting)  |        |        |             |            |             |            |                       |
| EU to EU                         | 8.546  | 5.486  | 1.805       | 1.200      | 2.150       | 1.370      | 1.232                 |
| AM to AM                         | 10.457 | 6.877  | 2.858       | 2.398      | 2.839       | 2.308      | 2.261                 |
| EU to AM                         | 8.538  | 5.479  | 1.767       | 1.172      | 2.116       | 1.350      | 1.094                 |
| (EU+AM) to AM                    | 9.605  | 6.261  | 2.614       | 2.319      | 2.618       | 2.323      | 2.248                 |
| Panel B: Same day (R-splitting)  |        |        |             |            |             |            |                       |
| EU to EU                         | 8.196  | 5.357  | 1.818       | 1.170      | 2.160       | 1.338      | 1.093                 |
| AM to AM                         | 9.961  | 6.678  | 2.818       | 2.298      | 2.789       | 2.233      | 2.213                 |
| EU to AM                         | 8.188  | 5.350  | 1.804       | 1.198      | 2.164       | 1.384      | 1.121                 |
| (EU+AM) to AM                    | 9.181  | 6.095  | 2.602       | 2.241      | 2.575       | 2.223      | 2.183                 |
| Panel C: One and Five days ahead |        |        |             |            |             |            |                       |
| EU to EU                         | 8.179  | 5.353  | 1.663       | 1.124      | 1.931       | 1.304      | 1.112                 |
| AM to AM                         | 9.891  | 6.648  | 2.685       | 2.275      | 2.600       | 2.184      | 2.172                 |
| EU to AM                         | 8.171  | 5.346  | 1.634       | 1.147      | 1.914       | 1.150      | 1.067                 |
| (EU+AM) to AM                    | 9.132  | 6.072  | 2.607       | 2.221      | 2.449       | 2.209      | 2.157                 |

*Note:* This table displays the in-sample RMSE (in %) for same-day, one- and five-days ahead predictions. The columns outline various models, encompassing the BS and Heston models fitted via calibration (columns 1 and 2) and then their versions implemented with neural networks, each subject to either an independent or sequential training methodology (columns 3 to 6). Column 7 presents the RMSE from the sequentially-trained Pure neural network. The rows categorize different training datasets employed for model fitting, with “EU”, “AM”, and “EU+AM” indicating the use of European, American, or a combination of both options data, respectively. Panel A details the in-sample RMSE for K-splitting, whereas Panel B relates to R-splitting; Panel C reports the one- and five-day-ahead in-sample RMSE. The period of study for this analysis extends from January 2016 to August 2023. For each setup, the model achieving the lowest RMSE is highlighted in bold, in all panels.

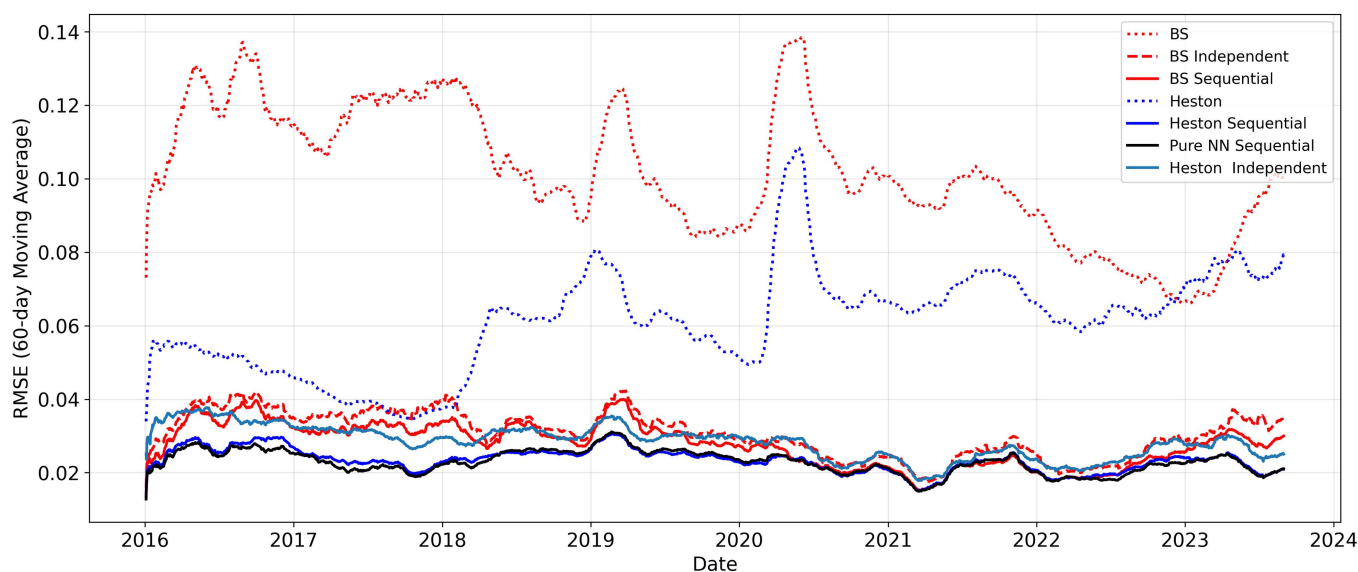
**TABLE 5** | Out-of-sample RMSE.

|                                 | BS     | Heston | BS+NN       |            | Heston+NN   |            | Pure NN    |
|---------------------------------|--------|--------|-------------|------------|-------------|------------|------------|
|                                 |        |        | Independent | Sequential | Independent | Sequential | Sequential |
| Panel A: Same day (K-splitting) |        |        |             |            |             |            |            |
| EU to EU                        | 6.890  | 4.930  | 1.791       | 1.317      | 2.136       | 1.278      | 1.084      |
| AM to AM                        | 7.920  | 5.909  | 2.569       | 2.209      | 2.485       | 1.926      | 1.842      |
| EU to AM                        | 8.003  | 5.941  | 3.644       | 3.347      | 3.900       | 3.163      | 2.982      |
| (EU+AM) to AM                   | 7.937  | 5.893  | 2.647       | 2.426      | 2.477       | 2.162      | 2.065      |
| Panel B: Same day (R-splitting) |        |        |             |            |             |            |            |
| EU to EU                        | 8.451  | 5.348  | 2.204       | 1.685      | 2.247       | 1.437      | 1.167      |
| AM to AM                        | 10.178 | 6.578  | 3.219       | 2.826      | 2.867       | 2.366      | 2.345      |
| EU to AM                        | 10.436 | 6.672  | 4.812       | 4.374      | 4.541       | 3.726      | 3.705      |
| (EU+AM) to AM                   | 10.286 | 6.603  | 3.432       | 3.122      | 2.895       | 2.560      | 2.510      |
| Panel C: One day ahead          |        |        |             |            |             |            |            |
| EU to EU                        | 8.583  | 5.422  | 3.904       | 2.692      | 4.012       | 2.515      | 2.554      |
| AM to AM                        | 10.446 | 6.709  | 4.051       | 3.408      | 3.519       | 2.927      | 2.901      |
| EU to AM                        | 10.720 | 6.810  | 5.762       | 4.803      | 5.096       | 4.728      | 4.073      |
| (EU + AM) to AM                 | 10.570 | 6.740  | 3.440       | 3.765      | 3.546       | 3.128      | 3.058      |
| Panel D: Five days ahead        |        |        |             |            |             |            |            |
| EU to EU                        | 8.683  | 5.609  | 4.058       | 3.500      | 4.242       | 3.166      | 3.244      |
| AM to AM                        | 10.516 | 6.862  | 4.813       | 4.104      | 5.341       | 3.766      | 3.680      |
| EU to AM                        | 10.785 | 6.957  | 5.828       | 5.357      | 5.719       | 4.686      | 4.592      |
| (EU + AM) to AM                 | 10.634 | 6.888  | 4.586       | 4.206      | 4.403       | 3.826      | 3.797      |

*Note:* This table displays the out-of-sample RMSE (in %) for same day, one-day- and five-days-ahead predictions. The columns outline various models, encompassing the BS and Heston models fitted via calibration (columns 1 and 2) and then their versions implemented with neural networks, each subject to either an independent or sequential training methodology (columns 3 to 6). Column 7 presents the RMSE from the pure neural network with sequential training. The rows categorize different training datasets employed for model fitting, with “EU,” “AM,” and “EU + AM” indicating the use of European, American, or a combination of both options data, respectively. Panel A details the out-of-sample RMSE for same-day predictions with K-splitting, whereas Panel B relates to R-splitting. Panel C reports the one-day-ahead out-of-sample RMSE, and Panel D presents the five-days-ahead out-of-sample RMSE. This analysis spans the sample period from January 2016 through August 2023. For each setup, the model achieving the lowest RMSE is highlighted in bold, in all panels.



**FIGURE 3** | 60-day moving average RMSE for same day predictions using K-splitting. This figure illustrates the 60-day moving average of the RMSE for same day predictions using K-splitting. The lines in the figure show the RMSE values obtained via traditional calibrations of the Black-Scholes and Heston models, as well as RMSE obtained from implementing these models via a neural network with independent and sequential training approaches, and from implementing a pure neural network with sequential training.



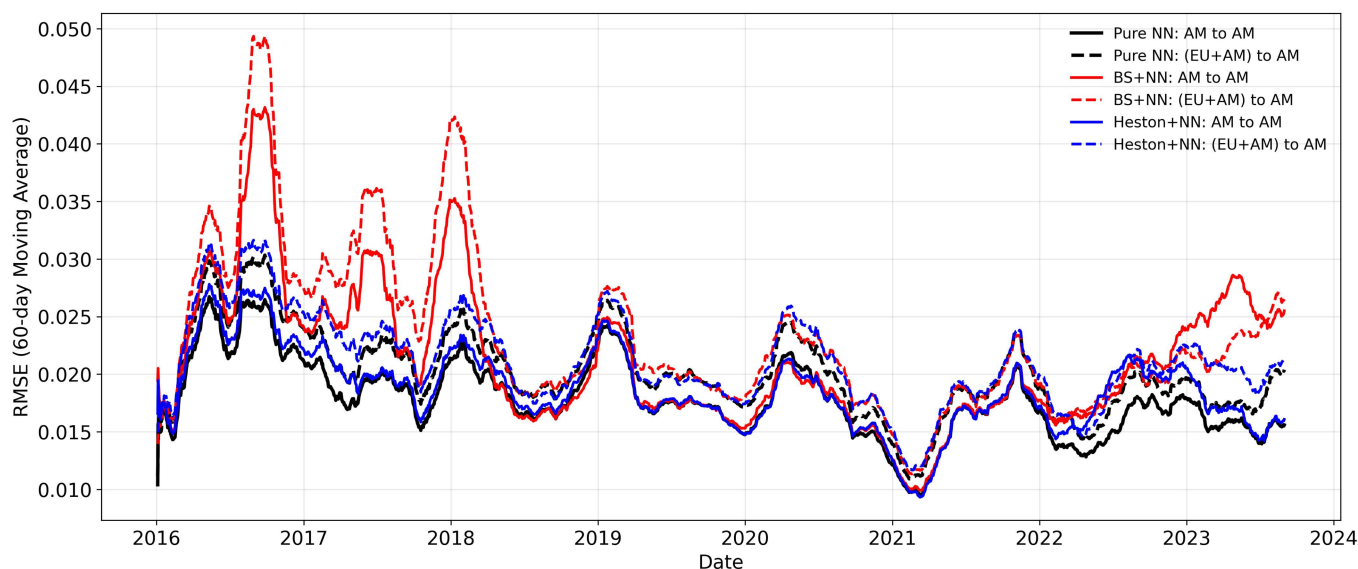
**FIGURE 4** | 60-day moving average RMSE for same day predictions using R-splitting. This figure illustrates the 60-day moving average of the RMSE for same day predictions using R-splitting. The lines in the figure show the RMSE values obtained via traditional calibrations of the Black-Scholes and Heston models, as well as RMSE obtained from implementing these models via a neural network with independent and sequential training approaches, and from implementing a pure neural network with sequential training.

achieve lower RMSE than the traditional Black-Scholes and Heston calibrations for same day predictions across both K-splitting and R-splitting methods (Panels A and B), as well as for 1-day-ahead and 5-days-ahead forecasts (Panels C and D, respectively). This indicates that incorporating neural networks substantially improves the accuracy of implied volatility fitting and forecasting over multiple horizons. Figures 3 and 4 visually reinforce this pattern, showing that neural network-based models maintain lower RMSE trajectories throughout the sample period compared to classical calibrated Black-Scholes and Heston models. In addition, while models boosted with neural network

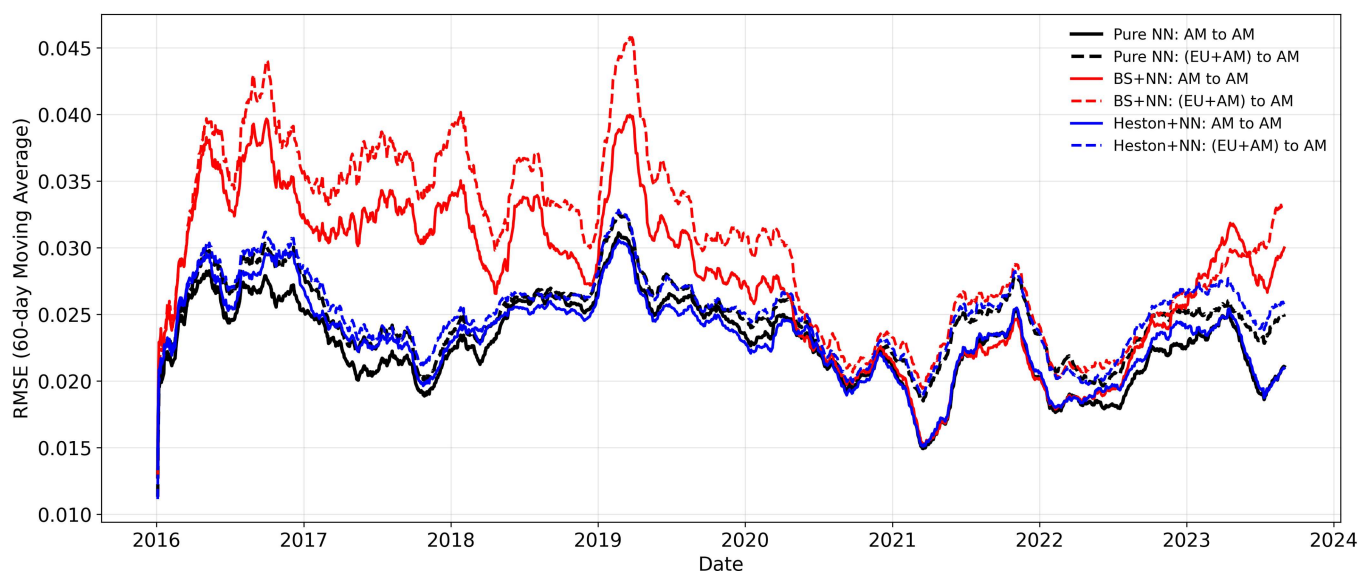
approaches perform well across different datasets and forecast horizons, the sequentially trained Pure NN generally delivers the lowest or near-lowest out-of-sample RMSE.

When comparing classical models without neural network enhancements, the Heston model consistently outperforms the Black-Scholes model across various configurations, as shown in Table 5. This result aligns with the theoretical advantage of the Heston model, which captures the stochastic nature of implied volatility, in contrast to the constant volatility assumption inherent in the Black-Scholes framework. A similar pattern emerges when these models are boosted with neural networks:





**FIGURE 5** | 60-day moving average RMSE for same day predictions using K-splitting, comparing different setups. This figure displays the 60-day moving average RMSE for same day predictions using K-splitting. The lines represent the RMSE values obtained from all different neural network frameworks (sequentially trained) via two prediction setups (AM to AM, and EU + AM to AM). The data sample covers the period from 2016 to 2023.



**FIGURE 6** | 60-day moving average RMSE for same day predictions using R-splitting, comparing different setups. This figure displays the 60-day moving average RMSE for same day predictions using R-splitting. The lines represent the RMSE values obtained from all different neural network frameworks (sequentially trained) via two prediction setups (AM to AM, and EU + AM to AM). The data sample covers the period from 2016 to 2023.

in forecasting American option implied volatilities, the Heston +NN specification generally achieves lower out-of-sample RMSE than the BS+NN counterpart, particularly under sequential training. This suggests that the richer volatility structure of the Heston model remains beneficial even when combined with flexible machine learning techniques.

A clear distinction in the out-of-sample results emerges between independent and sequential training approaches. Across all model specifications and forecast horizons, independently trained neural networks exhibit substantially higher RMSE than their sequentially trained counterparts, indicating

the weakness of independent training. In contrast, sequential training consistently improves forecasting performance by allowing the neural network to exploit temporal dependence in implied volatility dynamics. These gains persist across both K-splitting and R-splitting methods as well as across different forecast setups, highlighting the importance of incorporating time-series information when applying neural networks to implied volatility forecasting.

The RMSE values for same-day predictions are overall lower under K-splitting than under R-splitting, and the relative performance of the models/neural networks remains stable. In

particular, neural network-based specifications consistently outperform classical parametric models.

Table 5 shows that incorporating European option data when training neural networks to predict American option implied volatility yields mixed results. While the inclusion of European options does not uniformly improve forecasting performance, it can reduce RMSE for certain model configurations. For example, for same day K-splitting and 5-days-ahead forecasts, the “EU + AM to AM” Heston + NN model with independent training outperforms the corresponding “AM to AM” Heston + NN specification, suggesting that combining datasets can be beneficial under specific conditions. The reported RMSE magnitudes are economically reasonable considering the range of the implied volatilities, which is 4.12% to 124.86%. The magnitude of the RMSE values, as well as the relative reduction achieved by the neural network in our paper, for European options, is consistent with those reported in the literature, such as in Almeida et al. (2023), although our sample includes in-the-money options and options with longer maturities as well. As clearly illustrated in Figures 5 and 6, the RMSE for the “EU +

AM to AM” setup is consistently higher than that of the “AM to AM” setup under sequential training for all neural network frameworks, indicating that the advantage of incorporating European options does not persist when the training of neural networks exploits temporal dynamics.

Table 6 reports the out-of-sample RMSE for 1-day-ahead and 5-days-ahead implied volatility forecasts across option moneyness categories for the “AM to AM” prediction setup. Forecast errors are generally lower for ATM options than for ITM and OTM options across all models and forecasting horizons, consistent with the higher liquidity of ATM options. The models boosted with neural networks outperform the classical Black-Scholes and Heston calibrations across all moneyness categories, with particularly pronounced improvements for ITM and OTM options. Among the neural network specifications, the Pure NN and Heston+NN models achieve the lowest RMSE in most categories.

The results indicate that Pure NN has the best performance, suggesting that the BS+NN and Heston+NN models are likely incorrectly specified and cannot fully capture the behavior of

**TABLE 6** | RMSE across moneyness categories (ITM, ATM, OTM).

|                                        | Call         |              |              | Put          |              |              |
|----------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                                        | ITM          | ATM          | OTM          | ITM          | ATM          | OTM          |
| <b>Panel A: Same day (K-splitting)</b> |              |              |              |              |              |              |
| BS                                     | 11.528       | 5.350        | 6.588        | 5.698        | 4.829        | 7.778        |
| Heston                                 | 5.773        | 4.340        | 6.658        | 9.060        | 4.445        | 3.715        |
| BS+NN Sequential                       | 2.685        | 1.136        | 2.577        | 2.249        | 1.096        | 2.269        |
| Heston+NN Sequential                   | 2.052        | 0.728        | 2.357        | 2.334        | 0.764        | 2.055        |
| Pure NN Sequential                     | <b>1.970</b> | <b>0.702</b> | <b>2.335</b> | <b>2.145</b> | <b>0.632</b> | <b>1.992</b> |
| <b>Panel B: Same day (R-splitting)</b> |              |              |              |              |              |              |
| BS                                     | 14.863       | 4.967        | 5.899        | 5.921        | 4.563        | 10.796       |
| Heston                                 | 6.676        | 4.009        | 6.577        | 9.795        | 4.123        | 4.749        |
| BS+NN Sequential                       | 3.622        | 1.128        | 2.654        | 2.429        | 1.038        | 2.999        |
| Heston+NN Sequential                   | <b>2.407</b> | <b>0.786</b> | 2.640        | 2.519        | 0.804        | 2.743        |
| Pure NN Sequential                     | 2.456        | 0.805        | <b>2.571</b> | <b>2.420</b> | <b>0.732</b> | <b>2.725</b> |
| <b>Panel C: One day ahead</b>          |              |              |              |              |              |              |
| BS                                     | 15.254       | 4.990        | 5.983        | 5.959        | 4.583        | 10.889       |
| Heston                                 | 6.954        | 4.088        | 6.653        | 9.876        | 4.196        | 4.830        |
| BS+NN Sequential                       | 4.355        | 2.091        | 3.179        | 3.087        | 2.028        | 3.242        |
| Heston+NN Sequential                   | 3.285        | 1.824        | <b>2.954</b> | 3.058        | 1.828        | 2.992        |
| Pure NN Sequential                     | <b>3.276</b> | <b>1.824</b> | 2.978        | <b>3.023</b> | <b>1.777</b> | <b>2.921</b> |
| <b>Panel D: Five days ahead</b>        |              |              |              |              |              |              |
| BS                                     | 15.273       | 5.184        | 6.169        | 6.142        | 4.780        | 10.946       |
| Heston                                 | 7.028        | 4.350        | 6.922        | 10.043       | 4.453        | 4.979        |
| BS+NN Sequential                       | 4.868        | 3.240        | 3.972        | 3.784        | 3.152        | 3.981        |
| Heston+NN Sequential                   | <b>3.805</b> | 3.298        | 3.861        | 4.084        | 3.324        | 3.726        |
| Pure NN Sequential                     | 3.965        | <b>3.084</b> | <b>3.670</b> | <b>3.698</b> | <b>3.049</b> | <b>3.719</b> |

*Note:* This table presents the out-of-sample RMSE (in %) for the prediction of implied volatilities in the “AM to AM” setup, segmented by moneyness. It features four panels: Panel A corresponds to same-day K-splitting predictions, Panel B corresponds to same-day R-splitting predictions, Panel C corresponds to one-day-ahead predictions, and Panel D to five-days-ahead predictions. The rows detail five different models: the Black-Scholes and Heston models, each with and without enhancements from sequentially trained neural networks, and the sequentially trained pure neural network. The sample data spans from January 2016 to August 2023. Each column represents a moneyness category, with the lowest RMSE highlighted in bold, in all panels.

the data. Although forecasting accuracy declines at the 5-days horizon, the relative performance of the models remains stable across moneyness groups.

### 5.3 | Robustness

In this section, we perform a robustness analysis: instead of training the neural network on *all* available options, we use only actively traded options, defined as those with positive trading volume (volume > 0). While the full option dataset provides a richer cross-section for the implied volatility surface, the subset of traded-only options serves as a robustness test under conditions of extreme data sparsity. Whilst in this exercise we restrict the in-sample training set to traded options only, in the out-of-sample evaluation we use the same full option sample as in the baseline analysis, allowing for a direct comparison of forecasting performance.

Table 7 reports out-of-sample RMSE for same day, 1-day-ahead, and 5-days-ahead American option implied volatility forecasts where neural networks are trained exclusively on traded options. The results highlight several important findings. First, across all forecast horizons, for both K- and

R-splitting methods, and in all prediction setups, the sequentially trained pure neural network achieves the best overall performance, indicating that it has the strongest predictive accuracy even when the training data are severely restricted. This suggests that, under extreme data sparsity, the additional structure imposed by the traditional option pricing models (alone or boosted by neural networks) does not translate into improved out-of-sample performance. In addition, in this setting, hybrid specifications that combine neural networks with parametric models do not consistently outperform their traditional calibrations without neural networks.

Second, we observe that independently trained neural network specifications perform markedly worse than sequentially trained neural networks, and often also underperform the classical parametric models. This pattern indicates that independent training of neural networks is ill-suited to environments with extremely limited training data, such as the traded-only American options sample considered here.

Finally, a comparison of Table 7 with the out-of-sample results in Table 5 shows that the RMSE values are uniformly higher when the training set is restricted to traded options only. Since

**TABLE 7** | Robustness analysis: Out-of-sample RMSE.

|                                 |        |        | BS+NN       |            | Heston+NN   |            | Pure NN    |
|---------------------------------|--------|--------|-------------|------------|-------------|------------|------------|
|                                 | BS     | Heston | Independent | Sequential | Independent | Sequential | Sequential |
| Panel A: Same day (K-splitting) |        |        |             |            |             |            |            |
| EU to EU                        | 9.387  | 5.988  | 12.866      | 6.869      | 10.456      | 5.920      | 4.186      |
| AM to AM                        | 9.586  | 6.380  | 11.390      | 5.518      | 8.973       | 6.290      | 4.107      |
| EU to AM                        | 10.878 | 7.068  | 14.280      | 8.558      | 11.875      | 6.846      | 5.843      |
| (EU + AM) to AM                 | 9.564  | 6.387  | 11.199      | 5.640      | 9.099       | 5.857      | 3.564      |
| Panel B: Same day (R-splitting) |        |        |             |            |             |            |            |
| EU to EU                        | 11.042 | 6.483  | 41.172      | 11.137     | 36.505      | 6.983      | 6.684      |
| AM to AM                        | 12.378 | 7.294  | 35.915      | 8.290      | 32.878      | 7.923      | 5.772      |
| EU to AM                        | 13.076 | 7.630  | 39.615      | 12.117     | 36.813      | 8.781      | 8.195      |
| (EU + AM) to AM                 | 12.300 | 7.276  | 33.529      | 8.242      | 30.782      | 7.182      | 6.076      |
| Panel C: One day ahead          |        |        |             |            |             |            |            |
| EU to EU                        | 11.112 | 6.545  | 41.752      | 9.202      | 38.532      | 6.486      | 5.219      |
| AM to AM                        | 12.511 | 7.398  | 33.597      | 7.882      | 32.264      | 8.145      | 6.414      |
| EU to AM                        | 13.235 | 7.754  | 38.568      | 16.246     | 35.021      | 7.607      | 8.999      |
| (EU + AM) to AM                 | 12.498 | 7.398  | 31.802      | 8.032      | 30.344      | 7.895      | 5.552      |
| Panel D: Five days ahead        |        |        |             |            |             |            |            |
| EU to EU                        | 11.295 | 6.891  | 37.580      | 8.969      | 38.857      | 7.604      | 8.575      |
| AM to AM                        | 12.612 | 7.636  | 34.566      | 7.891      | 31.083      | 8.026      | 5.975      |
| EU to AM                        | 13.411 | 8.096  | 38.665      | 10.200     | 36.429      | 10.165     | 8.237      |
| (EU + AM) to AM                 | 12.588 | 7.618  | 32.052      | 7.882      | 28.772      | 7.816      | 6.370      |

*Note:* This table presents the RMSE (in %) for out-of-sample forecasts of implied volatilities when using traded option (volume > 0) as training set for same day-, one-day-, and five-days-ahead predictions. It features four panels: Panel A reports the out-of-sample RMSE for same-day forecasts using K-splitting, Panel B reports the corresponding results for R-splitting, Panel C presents the out-of-sample RMSE for one-day-ahead forecasts, and Panel D shows the results for five-day-ahead forecasts. The columns outline various models, encompassing the BS and Heston models fitted via calibration (columns 1 and 2), their versions implemented with neural networks, each subject to either an independent or sequential training methodology (columns 3–6), and the sequentially-trained pure neural network. The rows categorize different training datasets employed for model fitting, with “EU,” “AM,” and “EU + AM” indicating the use of European, American, or a combination of both options data, respectively. This analysis spans the sample period from January 2016 through August 2023. For each setup, the model achieving the lowest RMSE is highlighted in bold, in all panels.

the out-of-sample evaluation is identical across the two exercises, this difference reflects the reduced informational content of the traded-only training sample. Overall, the results indicate that using the full set of option prices, including options with zero trading volume, leads to more accurate option implied volatility forecasts.

For completeness, in Table A.1, we report the in-sample RMSE results when training is restricted to traded options. In Table A.2 of the Supplementary Appendix, we present the corresponding out-of-sample RMSE results when we forecast the implied volatilities only for traded option. In both cases, the main conclusions of the paper remain unchanged.

## 6 | Conclusions

This paper explores the use of neural networks and their integration into two classic option pricing models, the Black-Scholes and Heston models, to improve forecasting of American options implied volatilities. The study uses European and American options on the S&P 100 index from January 2016 to August 2023, to improve the performance of American option pricing models (via reducing the discrepancies between market-implied and model-implied volatilities) and to compare their forecast performance with that of a pure data-driven neural network framework. The findings of the study indicate that incorporating neural networks significantly reduces the RMSE of the pricing models, enhancing the models' accuracy to forecast implied volatilities.

In addition, we compare two neural network training strategies: independent daily training and sequential training that uses the network parameter set optimized on the previous day. Across all forecast horizons, option categories, and training data splitting methods, the sequentially trained pure neural network consistently delivers the best out-of-sample performance. This key finding highlights the importance of exploiting temporal dependence in implied volatility dynamics, as sequential training allows the network to efficiently incorporate information from the past. It also stresses that data-driven neural networks, when sequentially trained, can successfully improve forecasting of American option implied volatilities.

Our study also examines whether European option data provide additional useful information for forecasting American option implied volatility. The evidence is mixed: incorporating European options in the training set can improve forecasting performance under certain conditions, particularly for independently trained neural networks. Additionally, we document that training the neural networks on the full option cross-section, including options with zero trading volume, significantly improves predictive accuracy, highlighting the informational content of non-traded option quotes.

The application of neural networks in option pricing can markedly improve the prediction accuracy for American option implied volatilities at different forecasting horizons, and this finding represents a vital advancement for financial institutions and traders who depend on precise option pricing for portfolio management and strategic investment decisions. By implementing neural networks, option pricing models can better learn the non-linear dynamics and intricate patterns in data, thereby refining implied volatility forecasts and enabling more

accurate option pricing. Neural networks can incorporate comprehensive market information which enhances the overall accuracy of option pricing models.

## Author Contributions

**Haitong Jiang:** conceptualization (equal), data curation (equal), formal analysis, investigation, methodology (equal), project administration (equal), resources, software, validation (equal), visualization, writing - original draft preparation, writing - review and editing (equal). **Emese Lazar:** conceptualization (equal); data curation (equal), methodology (equal), project administration (equal), supervision (equal), validation (equal), writing - review and editing (equal). **Miriam Marra:** conceptualization (equal), data curation (equal), methodology (equal), project administration (equal), supervision (equal), validation (equal), writing - review and editing (equal).

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## Conflicts of Interest

The author declare no conflicts of interest.

## Data Availability Statement

The data used in this study were obtained from several key sources: OptionMetrics, accessed via the Wharton Research Data Services (WRDS) platform, Bloomberg, and the St. Louis Federal Reserve Economic Data (FRED) database. Restrictions apply to the availability of these data, which were used under license for the current study. Data are available with the permission of the respective data providers. The data that support the findings of this study are available from OptionMetrics. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from <https://optionmetrics.com/> with the permission of OptionMetrics.

## Endnotes

<sup>1</sup>In-sample and out-of-sample are used throughout this paper to refer to the training and testing datasets, respectively.

<sup>2</sup>Table 4 reports the in-sample results for the 1-day and 5-days ahead forecasts in a single panel (Panel C) because the out-of-sample results for these two horizons are based on the same in-sample training.

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### Supporting Information

Additional supporting information can be found online in the Supporting Information section.  
Supplementary Information